# SUMMARY OF ACCOMPLISHMENTS

**Name:** Sajjad Daroudi

**Program:** Mechanical Engineering Ph.D.

## POSITIONS/EMPLOYMENT

**Estimation/Robotics Researcher** at University of Minnesota Jan 2024- Present
**Research Assistantship, 50% Graduate Appointment**
**Laboratory for Innovations in Sensing, Estimation, and Control (LISEC)**
**Program:** Mechanical Engineering Ph.D.
**Advisor:** Dr. Rajesh Rajamani
**Project:** Estimation of end effector position in off-road vehicles using vehicle-level and actuator-level solutions.

- Estimation approach based on linkage angle estimation using IMUs for off-road vehicles in the presence of external disturbances.
- Estimation approach based on ferrous actuator piston displacement using magnetic sensors in the presence of magnetic hysteresis.
- Estimation approach based on development of a new type of ultrasonic sensor for two-dimensional end-effector planar position measurement.
- Design of nonlinear/high-gain observers for non-linear systems.

**Biomechanics Researcher** at Sharif University of Technology July 2021- July 2023
**Spine Biomechanics Laboratory**
**Program:** Mechanical Engineering MSc
**Advisor:** Dr. Navid Arjmand
**Project:** Estimation of biomechanical loads during load-lifting activities using deep-learning approach.

- Modeling load-lifting activities to obtain biomechanical loads.
- Developing multi-layer neural networks to estimate biomechanical loads.
- Developing a risk-assessment tool for spine injury in workers.

## HONORS AND AWARDS

*Winner of the "Mechanical Engineering Department **Fellowship Award"** from the University of Minnesota, valued at over $53,000, Fall 2023*
*Graduated from the **best-ranked University** in Iran with high GPAs:*

- *MSc degree from the Sharif University of Technology with a GPA of 18.96/20 (4/4).*
- *Automatic admission to MSc program of Sharif University of Technology for being among the top-ranked BSc students.*

- *BSc degree from Sharif University of Technology with a GPA of 17.4/20 (3.9/4). Sharif University of Technology is the best University in Iran based on the QS World University Rankings (2021, 2022, 2023, 2024) and 2024 Times Higher Education (THE) World University Ranking.*

**Teaching**

Teaching Assistant for "Fundamentals of Design and Manufacturing" at University of Minnesota (Fall 2023)

Teaching Assistant for "Numerical Computations" at Sharif Uni of Tech (Spring 2022)

**Convention for listing authorship:** *The most important technical contributor to the publication is the first author. ^ Indicates a publication with equal author contribution. The inventors listed on the patent have the same contribution to the invention. *Indicate which patent, paper, or presentation resulted from current doctoral research.*

## PUBLICATIONS

### Journal and Refereed Conference Articles

[1] ***Sajjad Daroudi**, Mike Gust, Rajesh Rajamani, "IMU-Based End-Effector Position Estimation on Agriculture and Construction Road Vehicles", IEEE Access *(Accepted for publication, Jan 2026)*

[2] ***Sajjad Daroudi**, Ali Zemouche, Rajesh Rajamani, "Observer Design for Highly Nonlinear Systems in the Presence of Hysteresis: Application to Long-Stroke Ferrous Actuator Position Estimation" *(Submitted to IEEE Transactions on Instrumentation and Measurement)*

[3] ***Sajjad Daroudi**, Rajesh Rajamani, "Magnetic Position Estimation in Ferrous Actuators Under Hysteresis: A Switched-Gain Non-Linear Observer Approach", Proceedings of the 2026 American Control Conference (*Decision expected by Jan 19, 2026)*

[4] ***Sajjad Daroudi**, Mike Gust, Rajesh Rajamani, "End-Effector Position Estimation on Off-Road Vehicles Using IMUs", Proceedings of the 2025 American Control Conference (ACC), pp. 4591-4596, IEEE, August 2025 *(Published)*

[5] **Sajjad Daroudi**, Navid Arjmand, Mahdi Mohseni, Marvan El-Rich, Mohamad Parnianpour, **"**Evaluation of ground reaction forces and centers of pressure predicted by AnyBody Modeling System during load reaching/handling activities and effects of the prediction errors on model-estimated spinal loads**"**, Journal of Biomechanics, Feb 2024 (*Published)*

[6] Mahziyar Darvishi, **Sajjad Daroudi**, Shahabedin Tavasoli, Ali Shafiezadeh, Farzam Farahmand, "Generalizability of motor modules across walking-based and in-place tasks – a distribution-based analysis on total knee replacement patients", Frontiers in Bioengineering and Biotechnology, Vol. 13, April 2025 *(Published)*

[7] Mahziyar Darvishi, Ali Shafiezadeh, **Sajjad Daroudi**, Shahabedin Tavasoli, Farzam Farahmand, "Kinematic modularity analysis of walking-based and in-place tasks", 2022 7th International Conference on Biomedical Engineering (ICBME), pp. 149-154, IEEE, March 2023 *(Published)*

**Patent**

[1] * Rajesh Rajamani and **Sajjad Daroudi,** "NON-REFLECTING SONAR POSITION SENSORS, SYSTEMS, AND METHODS OF USE FOR MEASURING ACTUATOR DISPLACEMENT", May 2025, Provisional Patent Application Filed by University of Minnesota *(Filed Patent, Co-Inventor)*

**PRESENTATIONS, POSTERS AND EXHIBITS**

**Invited Presentations at Professional Meetings, Conferences, etc.**

[1] * **Sajjad Daroudi**, Mike Gust, Rajesh Rajamani, "End-Effector Position Estimation on Off-Road Vehicles Using IMUs", 2025 American Control Conference (ACC), July 2025, Denver, CO *(Presented in invited session on "Mechatronics")*

**Abstracts** ***(published in electronic or paper format that are archived or searchable)***

[1] **Sajjad Daroudi**, Karim Ismail, Mahdi Mohseni, Marvan El-Rich, Navid Arjmand, "Accuracy of AnyBody Modeling System in predicting ground reaction forces and centers of pressure in lifting activities and effect of the prediction errors on spinal loads", 4th International Workshop on Spine Loading and Deformation, July 2023, *(Published Abstract)*

[2] Mahziyar Darvishi, **Sajjad Daroudi**, Shahabedin Tavasoli, Ali Shafiezadeh, Farzam Farahmand, Mina Baniasad, "Modular analysis of pooled muscle and kinematic data reveals the dynamics of the gait sub-phases", ESMAC 2022 Conference Abstracts, Sep 2022 *(Published Abstract)*.